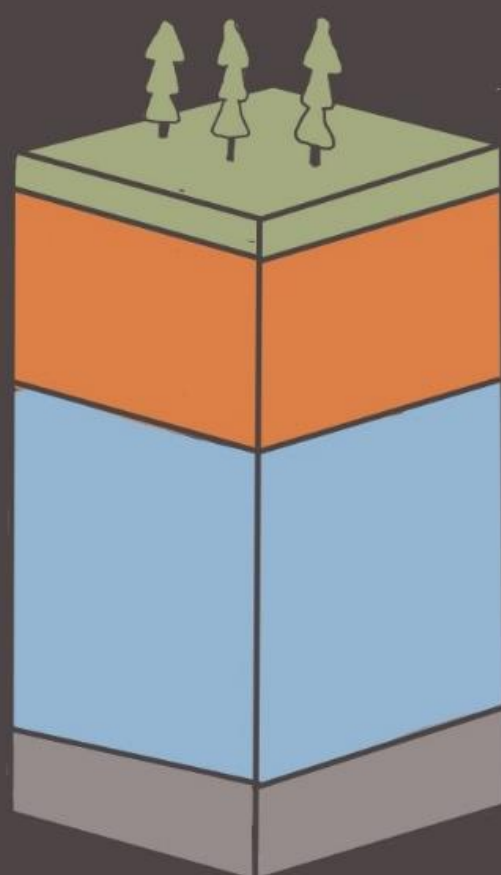


THE FUTURE OF PERMAFROST IN A CHANGING WORLD

WHAT IS PERMAFROST?

Permafrost is ground that has been at or below 0 degrees Celsius for two or more consecutive years [1].

Permafrost is underlain by **Unfrozen ground**.



Organic Materials such as vegetation or peat can cover the active layer, acting as insulation for the permafrost [2].

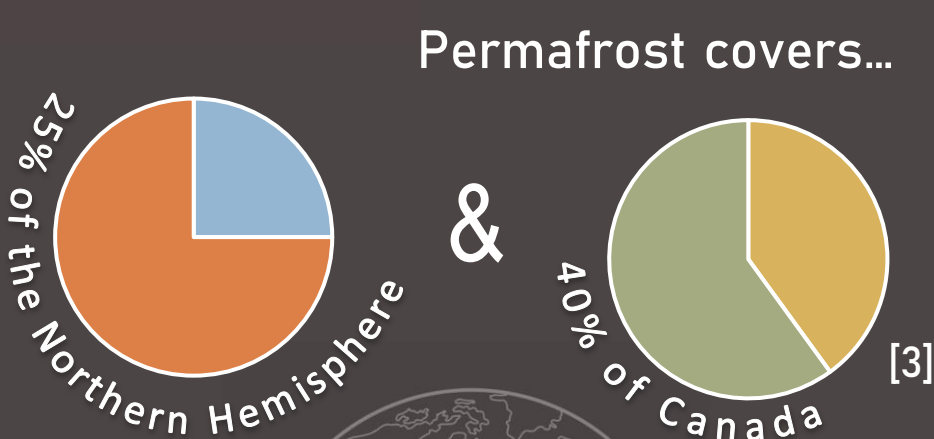
The Active Layer thaws and re freezes annually. It's thickness is dependent on the temperature of the ground surface, but can also be effected by overlying organic materials [2].



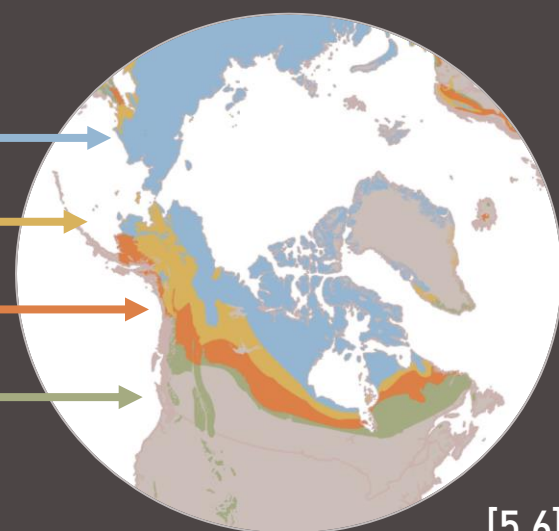
WHERE IS PERMAFROST?

Permafrost is found at high latitudes and altitudes[2].

Permafrost can be classified as ...



Continuous
Discontinuous
Sporadic
Isolated



... Depending on the percentage of ground permafrost covers [4].

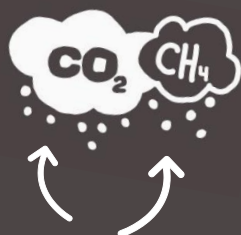
A THAWING WORLD



Global warming is resulting in permafrost warming, thaw and the thickening of the active layer [1].



In the high arctic, permafrost is increasing in temperature up to 0.5 degrees Celsius a decade [4].

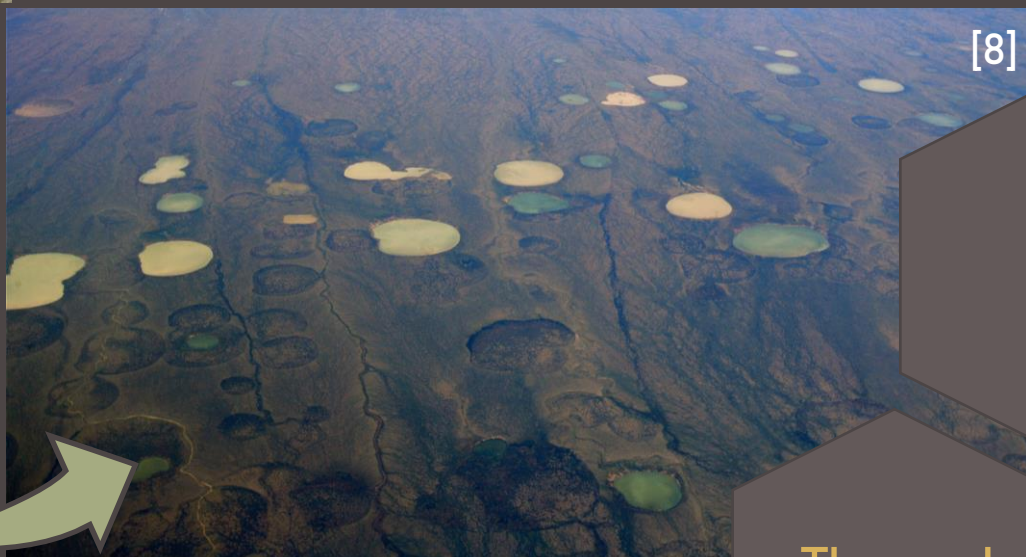


As permafrost thaws, GHG is released into the atmosphere [1].

The thawing of permafrost and the melting of ground ice can result in instability and erosion [2].



Melting ground ice can result in lake formation or collapses of surface features [1].



Thermal Erosion

HOW WILL THIS IMPACT CANADIANS?

Ground instability and subsidence threatens Canadians safety, and the long term viability of many communities[1].

Communities situated on continuous, ice rich permafrost will experience the most **HAZARDS AND DAMAGES** as Northern Canada warms [9].

Shifting buildings



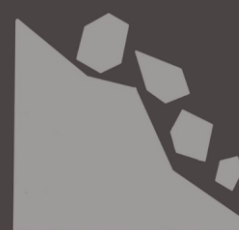
Cracked foundations

Expensive infrastructure damage

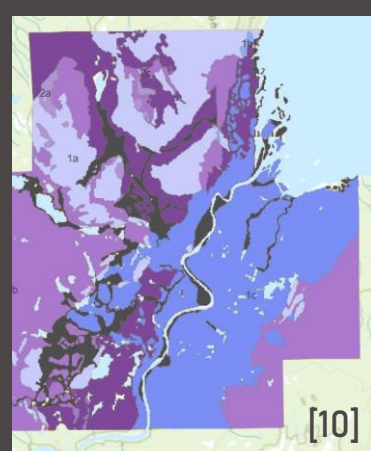
Damage to roads

Sinkholes

Increased risk of landslides

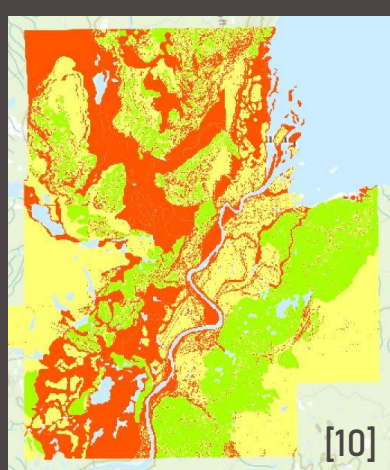


In vulnerable communities like **TASIUJAQ, QUEBEC** Hazard mapping and GIS are being used to aid in land use planning [9].



Risk factors in ground subsidence such as ice content, sediment type, slope, and drainage are mapped and ranked [9].

The final map identifies **high risk** land and is used in determining areas for safe future development [9].



Coastal communities experience coastal erosion up to 40m a year [11].

Want to learn more? Check out the references!

